ORGNZ-10: Advanced Segment Definitions

> **Series:** ORGNZ | **Notebook:** 10 of 10 | **Created:** February 2026 | **Last Updated:** 02/19/2026

Overview

This notebook is a deep-dive into the mechanics of creating effective segment definitions in Dynatrace Grail. While **ORGNZ-08** introduced segments, their structure, and design patterns, this notebook focuses on the practical details: filter condition syntax, data type include rules, metadata enrichment for segments, advanced variable patterns, and troubleshooting techniques.

The content draws from real-world implementation patterns, including host group-based segmentation strategies used in enterprise environments.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS environment with Grail enabled
Permissions	`storage:filter-segments:read` and `storage:filter-segments:write`
Knowledge	Completed ORGNZ-08 (Grail Segments fundamentals)
Data	At least 1 hour of log, span, and entity data

Learning Objectives

By the end of this notebook, you will:
- Master filter condition syntax including operators, field access, and combining conditions
- Understand data type include rules and the one-include-per-type constraint
- Know which metadata fields propagate across all signals (and which don't)
- Create advanced variable definitions using DQL entity queries
- Build host group-based segments from real-world naming conventions
- Troubleshoot common segment issues and apply performance best practices

1. Filter Condition Syntax

Segment filter conditions define which data a segment includes. Understanding the available operators and field access patterns is critical for building effective segments.

Supported Operators

Operator	Example	Notes
`=` (equals)	`k8s.namespace.name = "production"`	Exact match, best performance

Operator	Example	Notes
`!=` (not equals)	`k8s.namespace.name != "kube-system"`	Exclude specific values
`starts-with`	`dt.host_group.id = $platform*`	Prefix match with wildcard `*`
`IN`	`k8s.namespace.name IN ("prod", "staging")`	Match any value in list
`NOT IN`	`k8s.namespace.name NOT IN ("kube-system")`	Exclude list of values

> **Important — operator availability depends on data type:**
> - **Signal data includes** (logs, spans, events, metrics) support `=`, `!=`, `contains`, `not contains`, `starts with`, `ends with`, `IN`, and `NOT IN`.
> - **Classic entity includes** (`dt.entity.*`) only support `=` (with optional prefix wildcard `*`) and `in()`. `contains` and `ends-with` are **not available** for entity includes.
>
> For cross-signal consistency, prefer prefix-based naming conventions (e.g., `prod-web-tier` not `web-tier-prod`) so `starts-with` works everywhere.

Field Access Patterns

Pattern	Example	Use Case
Direct field	`k8s.namespace.name = "production"`	Primary Grail Fields
Tag matching	`matchesValue(tags, "env:production")`	Entity tags
Entity name	`entity.name = "my-service"`	Specific entity
Host group	`dt.host_group.id = "prod-web"`	Host group filtering (signal data)
JSON field	`content$.uri = "/health"`	Nested JSON in log content

Combining Conditions

- **AND** — Multiple conditions within a single include are AND-combined: `k8s.namespace.name = "production" AND k8s.cluster.name = "main-cluster"`
- **OR** — Use the `or` keyword within a single include (you cannot have multiple includes for the same data type): `k8s.namespace.name = "production" OR k8s.namespace.name = "staging"`

Test Your Filter Conditions

Before creating a segment, always validate your filter conditions with DQL. Run the equivalent filter clause as a DQL query to verify the results match your expectations.

> **Lab Exercise:** Complete Exercises 1-2 in **ORGNZ-10 LAB** for hands-on practice with these concepts.

2. Data Type Include Rules

Segments use **includes** to define which data types are filtered. Understanding how includes work is essential for building segments that behave as expected.

One Include Per Data Type

A segment can have **only one include block per data type**. You cannot define two separate include blocks for `logs`. To match multiple conditions, combine them with `OR` within a single include.

Supported Data Types

Data Type	Category	Example Filter
`logs`	Signal data	`k8s.namespace.name = "prod"`
`spans`	Signal data	`service.name starts-with "checkout"`
`events`	Signal data	`event.type = "CUSTOM_INFO"`

Data Type	Category	Example Filter
`bizevents`	Signal data	`event.provider = "my-app"`
`dt.entity.host`	Classic entity	`tags contains "env:prod"`
`dt.entity.service`	Classic entity	`entity.name starts-with "payment"`
`dt.entity.process_group`	Classic entity	`relationship: runsOn dt.entity.host`
`dt.entity.kubernetes_cluster`	Classic entity	`entity.name = "main-cluster"`

How Conditions Apply Per Data Type

When a segment is active and a user queries a specific data type, **only the matching include rule is injected**:

- Running `fetch logs` → only the `logs` include condition applies
- Running `fetch dt.entity.host` → only the `dt.entity.host` include condition applies
- Entity include rules do **NOT** filter signal data (and vice versa)

> **Critical:** Not all metadata is available on all data types. For example, `java.jar.file` exists only on spans. If you filter a segment by `java.jar.file`, it will only apply to spans — not to logs, metrics, or events. **Always use Primary Grail Fields for cross-signal segment filtering.**

Include Limits

Limit	Value
Maximum includes per segment	20
Include blocks per data type	1
Expressions per filter condition	10

Since each data type gets at most 1 include, a segment can cover up to 20 distinct data types or entity types.

3. Primary Grail Fields and Enrichment

The foundation of effective segment definitions is **metadata enrichment**. In Dynatrace's 3rd-generation architecture, each data point is treated independently. Unlike 2nd-gen where signals were tightly coupled to entities via Management Zones, in 3rd-gen every signal (logs, spans, metrics, events) must be enriched with the corresponding metadata.

Primary Grail Fields

Primary Grail Fields are infrastructure-related fields that are **automatically propagated across ALL data types** (spans, logs, metrics, topology). These are the ideal fields for segment filter conditions because they guarantee cross-signal consistency.

Primary Grail Field	DQL Field Name	Source
Host Group	`dt.host_group.id`	OneAgent host group configuration
Kubernetes Cluster	`k8s.cluster.name`	K8s integration
Kubernetes Namespace	`k8s.namespace.name`	K8s integration
AWS Account	`aws.account.id`	AWS integration
Azure Subscription	`azure.subscription`	Azure integration
GCP Project	`gcp.project.id`	GCP integration

> **Best Practice:** Always prefer Primary Grail Fields in segment definitions. They are indexed, automatically enriched, and consistent across all signal types.

Fields That Also Propagate to Service Metrics

Some fields go further and propagate to derived data like service metrics:

Field	Propagates to Service Metrics
`dt.security_context`	Yes
`dt.cost.costcenter`	Yes
`dt.cost.product`	Yes
Host group (`dt.host_group.id`)	Yes
Other host tags	**No**

Primary Grail Tags

When Primary Grail Fields don't align with how your organization wants to segment data (e.g., shared infrastructure with multiple apps on one host group), use **Primary Grail Tags**.

- Identified by the prefix `primary_tags.`
- Example: `primary_tags.stage=prod` , `primary_tags.team=platform`
- Set via host properties, environment variables (`DT_TAGS`), or OpenPipeline
- Propagated across all data points, similar to Primary Grail Fields

> **Limitation:** Primary Grail Tags do not yet enrich Kubernetes metrics or events. They work for logs, spans, and topology data. Check the Dynatrace documentation for the latest supported signal types.

Enrichment Approaches

Scenario	Approach	Effort	Granularity
Dedicated Infrastructure	Host group + host properties	Low	Host-level
Shared Infrastructure	`DT_TAGS` environment variable per process	Higher	Process-level
Kubernetes	Primary Grail Fields (automatic)	None	Namespace/cluster

Scenario	Approach	Effort	Granularity
Cloud (AWS/Azure/GCP)	Native cloud tags	None	Account/subscription

Enrichment via Host Properties

For dedicated infrastructure, enrich hosts via **Deployment Status** → select hosts → **Modify host properties**:

1. `dt.security_context=<team-or-app>` — For IAM access control
2. `dt.cost.costcenter=<cost-center>` — For cost allocation
3. `dt.cost.product=<product>` — For product tracking
4. `primary_tags.<key>=<value>` — For custom enrichment

> **Restart Requirements:**
> - **Logs**: No application restart needed — OneAgent log module auto-enriches after its own restart
> - **Spans**: Application restart **required** to apply enrichment to traces
> - **Metrics**: OneAgent restart applies automatically

Verify Enrichment Coverage

Before building segments, verify that the fields you plan to filter on actually exist on your signal data. Fields with 0% coverage need enrichment via OneAgent configuration or OpenPipeline.

> **Lab Exercise:** Complete Exercises 3-4 in **ORGNZ-10 LAB** for hands-on practice with these concepts.

4. Advanced Variable Patterns

Variables make segments dynamic by letting users select values at runtime (e.g., choosing a host group or namespace from a dropdown).

Primary vs Secondary Variables

The variable definition is a DQL query. The columns in the result set determine the variable behavior:

Column Position	Role	Behavior
First column	Primary variable	Shown in the dropdown selector
Additional columns	Secondary variables	Available for use in filter conditions

Variable Rules

Rule	Detail
Maximum values in dropdown	10,000 per variable
Selected values per segment	100 maximum
Wildcard	Allowed **after** variable name: `$cluster*` (starts-with)
No wildcard in names	Variable names and values cannot contain `*`
Permissions	Users must have read access to entities queried by the variable DQL
Empty dropdown	Usually means the user lacks permission for the variable query

Variable DQL Examples

The following queries demonstrate how to create variable definitions for common use cases.

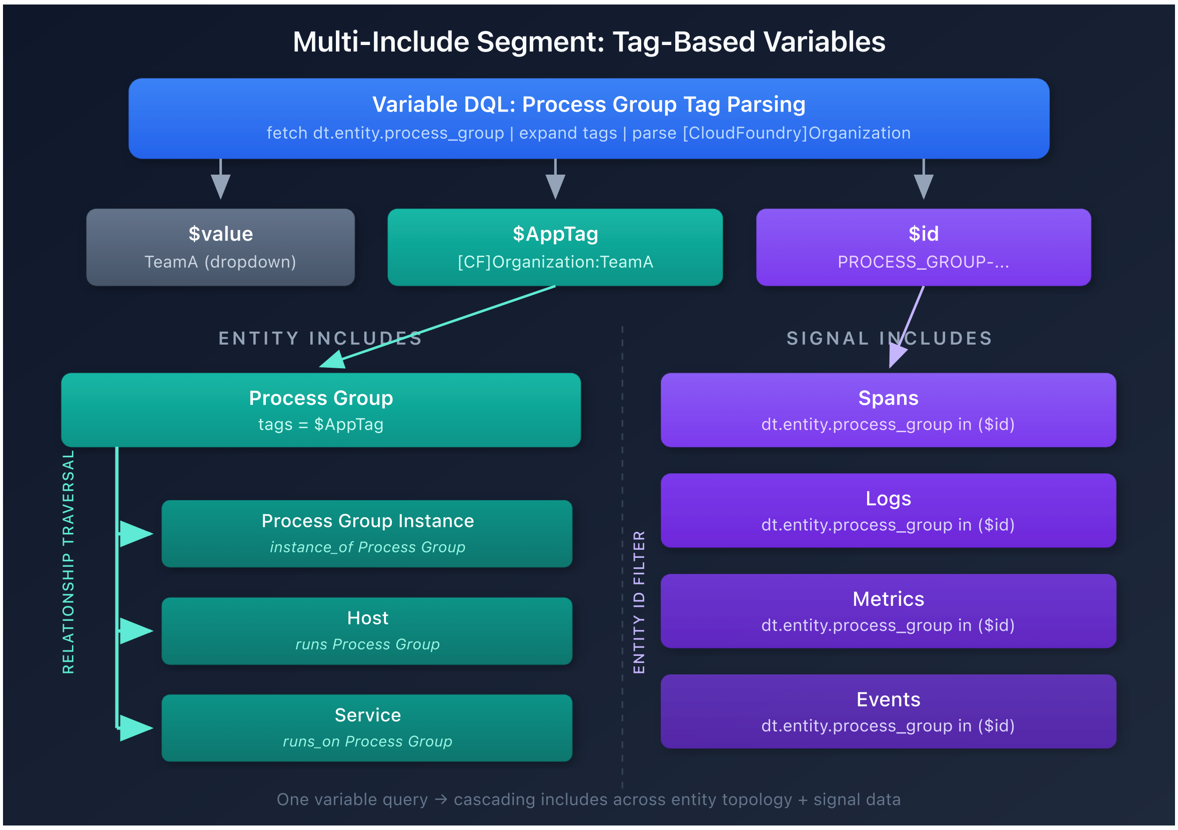
> **Lab Exercise:** Complete Exercises 5-6 in **ORGNZ-10 LAB** for hands-on practice with these concepts.

Multi-Include Segment Pattern: Tag-Based Variables

The CloudFoundry query above creates three variables from process group tags:

Variable	Example Value	Use
`$value`	`TeamA`	Display in dropdown
`$id`	`PROCESS_GROUP-1234567`	Filter signal data by entity ID
`$AppTag`	`[CloudFoundry]Organization:TeamA`	Match entity tags

Use these variables across **multiple includes** to build a segment that filters both entity and signal data:



Include Type	Filter	How It Works
Process Group	`tags = $AppTag`	Matches process groups by tag
Process Group Instance	`instance_of Process Group`	Relationship traversal — includes instances of matched process groups

Include Type	Filter	How It Works
Host	`runs Process Group`	Relationship traversal — includes hosts running matched process groups
Service	`runs_on Process Group`	Relationship traversal — includes services on matched process groups
Spans	`dt.entity.process_group in ($id)`	Signal include — filters spans by process group entity ID

> **Key Insight:** This pattern bridges entity and signal data in a single segment. Entity includes use tag matching and relationship traversal to cascade across the topology, while signal includes use the entity ID variable to filter spans, logs, and metrics directly. This is especially powerful when Primary Grail Fields are not available for your use case and you need to segment by application-level metadata like CloudFoundry organizations or custom tags.

Using Variables in Filter Conditions

After defining a variable query, reference the columns in your segment filter:

```
Filter: dt.host_group.id = $host_group OR dt.entity.host_group = $id
```

Where `$host_group` references the primary variable (first column) and `$id` references the secondary variable (second column).

Wildcard with variables — Use `$variable*` to match prefixes:

```
Filter: dt.host_group.id = $platform*
```

This matches all host groups that **start with** the selected platform value.

5. Host Group-Based Segments

A common enterprise pattern is encoding multiple dimensions into host group names using a naming convention. This section shows how to create segments from these encoded dimensions.

Real-World Pattern

Consider a customer using the host group naming convention `<platform>_<app>_<stage>`:

Host Group	Platform	App	Stage
`k8s_multi_prod`	k8s	multi	prod
`k8s_easytravel_staging`	k8s	easytravel	staging
`onPrem_easytravel_staging`	onPrem	easytravel	staging
`onPrem_easytrade_prod`	onPrem	easytrade	prod

Best Practice: One Segment Per Dimension

Create a **separate segment for each dimension** (platform, app, stage). This gives users the flexibility to combine them as needed.

Step 1: Extract Dimensions with DQL

Use DQL to parse the host group name and extract each dimension as a variable:

> **Lab Exercise:** Complete Exercise 9 in **ORGNZ-10 LAB** for hands-on practice with this concept.

Step 2: Create Variable DQL for Each Dimension

Platform variable:

```
fetch dt.entity.host_group
| parse entity.name, """LD:platform '_' LD:app '_' LD:stage"""
| dedup platform
| fields platform
```

App variable:

```
fetch dt.entity.host_group
| parse entity.name, """LD:platform '_' LD:app '_' LD:stage"""
| dedup app
| fields app
```

Stage variable:

```
fetch dt.entity.host_group
| parse entity.name, """LD:platform '_' LD:app '_' LD:stage"""
| dedup stage
| fields stage
```

Step 3: Define Filter Conditions

Segment	Filter Condition	Notes
Platform	`dt.host_group.id = $platform*`	Matches host groups starting with selected platform
App	`dt.host_group.id contains $app`	Matches host groups containing selected app (signal includes only)
Stage	`dt.host_group.id ends-with _$stage`	Matches host groups ending with selected stage (signal includes only)

> **Note:** `contains` and `ends-with` operators are available for signal data includes (logs, spans, events, metrics) but **not** for classic entity includes. For entity includes, use `starts-with` (prefix wildcard) patterns. Always test with the segment preview before deploying.

Step 4: Validate Across Signal Types

After creating the segments, validate they filter correctly across:
- **Logs** — Check record count reduction when segment is applied
- **Distributed Traces** — Verify spans are filtered by the host group dimension
- **Hosts / Services** — Confirm entity filtering matches expectations
- **Infrastructure** — Verify metrics are scoped correctly

> **Tip:** The host group is a Primary Grail Field, so it propagates across all signal types. This makes it an excellent candidate for segment filtering.

6. Segment Visibility and Sharing

Visibility Settings

Setting	Behavior
Unlisted (default)	Only visible to the creator and users it's shared with
Public	Visible in everyone's segment list in the environment

Permissions Model

Permission	Action	Included In
`storage:filter-segments:read`	View and use segments	Dynatrace Standard User
`storage:filter-segments:write`	Create and edit segments	Dynatrace Standard User
`storage:filter-segments:share`	Share with others	Dynatrace Standard User
`storage:filter-segments:delete`	Delete segments	Dynatrace Standard User
`storage:filter-segments:admin`	Manage segment permissions	Dynatrace Professional User

Governance Recommendations

- **Platform team** owns environment and infrastructure segments (public)
- **Application teams** own app-specific segments (shared with their group)
- Maintain a small set of "official" public segments
- Use consistent naming: `team-`, `env-`, `app-`, `region-` prefixes

7. Cross-App Integration

Cross-App Persistence

When you select a segment in one Dynatrace app (e.g., Logs), the selection **persists** as you navigate to other apps (e.g., Distributed Traces, Problems). This means you don't need to reapply filters when drilling into different data types during an investigation.

App Support Matrix

App	Segment Support	Notes
Dashboards	Full	Dashboard-level and tile-level segments
Notebooks	Full	Segment selector at top of notebook
Logs	Full	Filters all log queries
Distributed Traces	Full	Filters span queries
Problems	Full	Requires events include with `event.kind = "DAVIS_PROBLEM"`
SLOs	Full	Filters SLO evaluation scope
Site Reliability Guardian	Full	Filters validation scope
Workflows	Full	Available in workflow DQL steps
Smartscape	Full	Filters topology view

Dashboard Segment Behavior

- **Dashboard-level segment** applies to all tiles
- **Tile-level segment** overrides the dashboard segment for that specific tile
- Use this pattern for executive dashboards that show cross-team metrics alongside team-specific views

8. Known Limitations and Workarounds

Limitation	Detail	Workaround
detected problems require event includes	To filter problems with segments, define an events include with `event.kind = "DAVIS_PROBLEM"`; entity includes alone do not filter problem records	Add event-type includes alongside entity includes
Single relationship traversal	Entity relationships can only traverse one hop	Target specific entity types directly in includes
Inclusions only, no exclusions	Cannot say "everything EXCEPT team-X"	Explicitly include what you want (MZ supported exclusions; segments do not)
Entity includes: no contains/ends-with	Classic entity includes only support `=` and prefix wildcards; `contains` and `ends-with` are not available	Use prefix-based naming conventions; signal includes support broader operators
1 include per data type	Cannot have two `logs` include blocks	Combine conditions with `OR` within a single include
Max 20 includes per segment	Hard limit on rule count	Consolidate rules; use variables for flexibility
Max 10 expressions per filter	Limits filter complexity	Use OpenPipeline to pre-enrich data with simpler filter fields
Max 10 segments per query	Cannot stack unlimited segments	Design broader segments instead of many narrow ones
Variable dropdown empty	Users without entity read permissions see empty dropdown	Ensure variable DQL targets entities the user can read

9. Troubleshooting and Performance

Common Issues

Symptom	Likely Cause	Fix
Empty variable dropdown	Missing entity read permissions	Grant `dt.entity.*:read` to user's policy
Segment shows no data	Filter field not enriched on signal data	Verify enrichment with audit queries from Section 3
Segment filters entities but not logs	Entity includes don't apply to signals	Add separate signal-type includes (logs, spans)
Unexpected data included	OR logic combining values	Review filter condition logic within the include
Segment missing in app	Segment set to unlisted and not shared	Share segment or set visibility to public
Segment works for logs but not spans	Field exists on logs but not spans	Use Primary Grail Fields for cross-signal consistency

Performance Best Practices

Every segment filter condition is injected into every query the user runs. Keep conditions simple:

1. **Prefer Primary Grail Fields** — They are indexed and optimized for filtering
2. **Use exact matches** — `=` is faster than `starts-with`
3. **Minimize expressions** — Fewer conditions = better query performance
4. **Use OpenPipeline pre-processing** — If conditions would be complex, pre-enrich a simple field (like `dt.security_context`) and filter on that instead
5. **Narrow scope first** — Start with the most restrictive condition in AND combinations

Naming Conventions

Prefix	Use Case	Examples
`team-`	Team-based segments	`team-platform-infra`, `team-checkout`

Prefix	Use Case	Examples
`env-`	Environment segments	`env-production` , `env-staging`
`app-`	Application segments	`app-checkout-full` , `app-payments`
`region-`	Regional segments	`region-us-east` , `region-eu-west`

Segment Lifecycle

1. **Design** — Identify data types, filter conditions, variables
2. **Test** — Validate filter logic with DQL queries
3. **Create** — Build in the Segments app
4. **Share** — Distribute to appropriate users/groups
5. **Monitor** — Check for empty results, permission issues
6. **Retire** — Remove when no longer needed; review quarterly

Discover Available Filter Candidates

> **Lab Exercise:** Complete Exercises 10-11 in **ORGNZ-10 LAB** for hands-on practice with these concepts.

Summary

In this notebook you learned:

1. **Filter condition syntax** — Operators (`=` , `!=` , `starts-with` , `contains`), field access, AND/OR logic
2. **Data type include rules** — One include per type, conditions scoped to queried data type
3. **Primary Grail Fields** — Which metadata propagates across all signals (and which doesn't)
4. **Enrichment approaches** — Dedicated vs shared infrastructure, host properties vs DT_TAGS
5. **Advanced variables** — Primary/secondary, DQL queries, permission requirements

6. **Host group-based segments** — Parsing naming conventions, one segment per dimension
7. **Visibility and sharing** — Public vs unlisted, governance model
8. **Cross-app integration** — Persistence, dashboard vs tile-level segments
9. **Limitations and troubleshooting** — detected problem event includes, entity operator restrictions, performance tips

Series Summary

Notebook	Topic	Key Takeaway
ORGNZ-01	Introduction	Three pillars of data organization
ORGNZ-02	Grail Buckets	Bucket fundamentals and limits
ORGNZ-03	Bucket Strategy	Naming, retention, design patterns
ORGNZ-04	Permissions Overview	Permission levels and policy structure
ORGNZ-05	Bucket-Level Access	IAM policies for bucket isolation
ORGNZ-06	Security Context	Fine-grained access with dt.security_context
ORGNZ-07	Advanced Permissions	Record and field-level patterns
ORGNZ-08	Grail Segments	Segment fundamentals and design patterns
ORGNZ-09	Enterprise Patterns	Combined approaches at scale
ORGNZ-10	**Advanced Segment Definitions**	**Filter syntax, enrichment, variables, troubleshooting**

References

- Grail Segments
- Include Data in Segments
- Variables in Segments
- Segment Limits
- Segments in DQL Queries
- Supported Data Types in Segments
- Segments Blog: Cut Through the Noise
- Segments Blog: Empower Centralized Teams

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